



RAM

Whether it is in terms of size or bandwidth, memory can spell a world of difference in the response time of your applications

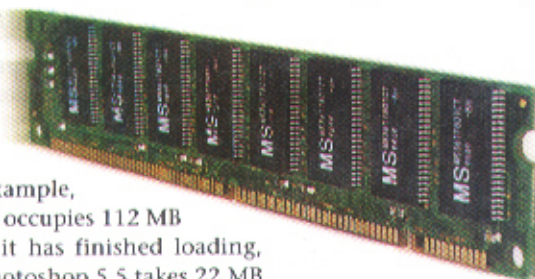
The centre stage for all your programs, RAM is where your OS' core and other applications are loaded. The unsaid rule of thumb states that the more RAM there is, the happier your programs are, giving you a smoother computing experience.

When you launch a program from your hard disk, a large part of it is loaded in RAM itself. The executable code occupies space in RAM with the amount varying from one application to another. For example, image editing software heavily depends on RAM for holding image files temporarily while you work on them. This slows down the other programs. For example, Windows 98 SE occupies 112 MB of RAM when it has finished loading, while Adobe Photoshop 5.5 takes 22 MB with no image files opened! Therefore, if you open too many applications simultaneously, after a certain point, all the useable part of the RAM will be utilised. After this, the temporary information gets saved on the hard disk, drastically slowing down your system since hard disks transfer data much slower than RAM. Therefore, if you are used to running many applications simultaneously, the more RAM you have, the smoother your system will run.

RAM is packaged in various densities, ranging from 64 MB, all the way to 512 MB per module. One bottleneck faced if you choose two modules of smaller size over one large module is the utilisation of precious RAM slots on your motherboard. This will hinder future upgrades

since you would have no more free slots.

Over the past two years, various types of RAM based on newer technologies have cropped up. The most popular RAM is SDRAM (Synchronous DRAM). In its 133 MHz version, it sports a data transfer speed of 1.064 GBps. This has since been superseded by faster technologies such as RDRAM and DDR SDRAM. For example, PC800 RDRAM has a data transfer speed of up to 1.6 GBps while PC2100 DDR SDRAM boasts of a transfer rate of 2.1 GBps. However, the type of memory you have or can

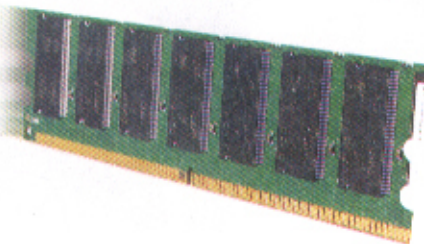


SDRAM is a dated technology and can slow down today's applications

upgrade to is also dictated by the type of motherboard/chipset combination you have—not all motherboards support all types of RAM interchangeably. There are physical differences in their form factor and construction, due to which they can only accept RAM modules of one type. For example, the i850 motherboards from Intel are based upon RDRAM while their i845 and i845D motherboards are based on SDRAM and DDR SDRAM respectively. Even though RAM is close to a thousand times faster than your hard disk drive and most other components in your system, the one component that outpaces RAM in terms of data

handling bandwidth is the processor. And it is this very component to which RAM is practically married! Therefore, if you are running any of the new processors such as the 2 GHz+ Pentium 4s or AMD's

Thoroughbred processors, it would make



DDR is going through many iterations and is fast becoming an industry standard

sense to pair them with the fastest RAM. This would be RDRAM with Intel's P4s and DDR SDRAM with AMD's Thoroughbreds.

Finally, an often-missed specification, known as Column Access Strobe (CAS), is another potential bottleneck. CAS defines the maximum time taken (in clock cycles) for the RAM to reach a piece of data inside it before it is ready for transfer. Imagine trying to locate a shop that sells magazines. CAS is akin to the time taken to reach this shop—once you're inside the shop, it would take you less time to take a set of magazines out of the shop (the RAM's data transfer rate). For example, 133 MHz SDRAM is available in three CAS flavours—2, 2.5 and 3 clock cycles. Naturally, the lower the time taken to access the data inside it, the quicker is the response of the RAM.

Tips & Tweaks

- Buy more RAM. Today's applications and operating systems such as Windows XP need at least 256 MB of RAM to run smoothly.
- From the BIOS, enable the lowest CAS setting that your RAM supports. Also set it at the highest bus speed setting of the RAM you are using—change it to 133 MHz if you have such RAM.
- Enable bank interleaving from the BIOS if your motherboard supports it.
- Close any large applications you do not need so as to conserve memory for other important programs.



RDRAM represents the best you can get in bandwidth, but its high latency is still a problem